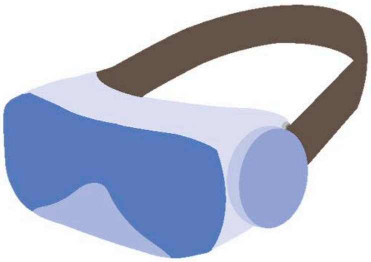
IN DEPTH

3D movies

In the 1890s, the first 3D movies were made using a technique called anaglyph 3D. Every scene was filmed with two cameras set a small distance apart. The two videos, one tinted blue and one tinted red, were combined digitally. Viewers wore glasses with one blue lens (to cancel out the red image) and one red lens (to cancel out the blue image). The result was a 3D-like effect.

Equipment

Rendering a visual display in a game or simulator requires a powerful computer to process thousands of numbers every second in order to decide what pixels to display. A graphics processing unit is a specialized processor designed to handle high volumes of basic maths operations. In other words, they work faster than a standard CPU, but they can only do certain types of work. If the GPU isn't powerful enough, the display will lag, creating a choppy, unrealistic experience.



△ **VR gear**
VR headsets cover the eyes. They render different images for each eye. Sensors inside the headset change the display in response to the user's head movements.

△ **Body suit**
These suits give tactile feedback through a mesh of sensors inside each suit's fabric. They can produce vibrations and simulate touching objects. Body suits offer users another layer of immersion into a virtual world.

The future of VR

For years, VR simulators have been used to train soldiers, policemen, and doctors as VR makes training safer, cheaper, and more effective. As the technology becomes cheaper and more accessible, high-quality simulators will be created for a wider range of jobs. Maybe someday VR will also include smells and physical sensations!

REAL WORLD

VR sickness

You've spent your entire life in the real world and are used to navigating it. Even the tiniest bit of lag on a headset display can cause a type of nausea known as virtual reality sickness. Another possible cause is related to motion sickness – where people experience nausea in response to movement – though people do not actually have to be moving to feel VR sickness.

△ **Playing games**
VR allows artists and game designers to take creativity to a new level. Instead of watching a screen, people can experience car chases and explore medieval castles in person.

△ **Education**
Virtual classrooms, where people can virtually experience training or a lesson, are possible with VR. The technology may also help with visualizing concepts in maths and physics.

△ **Travel experience**
Some companies offer tours of famous places and monuments through VR technology. People can explore different corners of the world without leaving their seats.